

SAFETY DATA SHEET

(SDS)

This safety data sheet complies with the requirements of:
Regulation (EC) No. 1907/2006 and Regulation (EC) No. 1272/2008, (EU) No. 2015/830

Revision Date 18-Apr-2024

WAI2 - EGHS - EUROPEAN

Revision Number 3

SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product Identifier

Product Name CISA™ A Component

Product No 941709 A
Unique Formula Identifier (UFI) Not applicable

REACH registration number Not applicable

Pure substance/mixture Mixture

Contains Nitric Acid

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Use as laboratory reagent

Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Manufacturer, Importer, Supplier Thermo Fisher Scientific©
Water and Lab Products
22 Alpha Road
Chelmsford, MA 01824, USA
1-978-232-6000

E-mail address wlp.techsupport@thermofisher.com

Made in USA

1.4. Emergency telephone number 24 Hour Emergency Phone Number
CHEMTREC®
Within USA and Canada: 1-800-424-9300
Outside USA and Canada: 1-703-527-3887
(collect calls accepted)

SECTION 2. HAZARDS IDENTIFICATION**2.1. Classification of the substance or mixture**

Classification - Mixture

Classification according to Regulation (EC) No. 1272/2008 [CLP]

Skin Corrosion/Irritation	Category 1 Sub-category B - (H314)
Serious eye damage/eye irritation	Category 1 - (H318)

2.2. Label elements

Contains Nitric Acid

**Signal Word**

Danger

Hazard Statements

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

Precautionary Statements

P264 - Wash face, hands and any exposed skin thoroughly after handling

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor/physician

P304 + P340 - IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P363 - Wash contaminated clothing before reuse

P501 - Dispose of contents/ container to an approved waste disposal plant

2.3. Other hazards**General Hazards**

This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

Component	EC No	CAS No	Weight %	CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567	REACH Reg. No
Water	EEC No. 231-791-2	7732-18-5	90 - 100%	Not classified	No information available
Nitric Acid	EEC No. 231-714-2	7697-37-2	0 - 10%	Ox. Liq. 2 (H272) Acute Tox. 1 (H330) Skin Corr. 1A (H314) (EUH071)	No information available

Component	CAS No	Specific concentration limits (SCL's)	M-Factor	Component notes
Water	7732-18-5	-	-	-
Nitric Acid	7697-37-2	Ox. Liq. 2 (H272) :: C>=99% Ox. Liq. 3 (H272) :: 70%<=C<99% Ox. Liq. 3 (H272) :: C>=65% Skin Corr. 1A (H314) :: C>=20% Skin Corr. 1B (H314) :: >=5%<-C<20%	-	-

Component	ECHA (RAC) ATE (Oral)	ECHA (RAC) ATE (Dermal)	ECHA (RAC) ATE (Inhalation)
Nitric Acid	-	-	ATE = 2.65 mg/L (vapour)

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.
Inhalation	If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.
Ingestion	Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.
Self-Protection of the First Aider	Use personal protective equipment as required. See section 8 for more information. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

4.2. Most important symptoms and effects, both acute and delayed

Most important symptoms and effects	Causes burns by all exposure routes
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4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically
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SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
CO₂, dry chemical, dry sand, alcohol-resistant foam.

Unsuitable Extinguishing Media
No information available

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

6.2. Environmental precautions**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

6.3. Methods and material for containment and cleaning up**Methods for Containment**

Prevent further leakage or spillage if safe to do so.

Methods for Cleaning Up

Soak up with inert absorbent material. Pick up and transfer to properly labeled containers.

Reference to Other Sections

Refer to protective measures listed in Sections 7 and 8

See Section 8 for information on appropriate personal protective equipment

See Section 12 for additional Ecological Information

See Section 13 for additional waste treatment information

SECTION 7: HANDLING AND STORAGE**7.1. Precautions for safe handling****Advice on safe handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

General hygiene considerations

Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities**Storage Conditions**

Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place.

7.3. Specific end use(s)**Specific Use(s)**

Use as laboratory reagent

Risk Management Methods (RMM)

The information required is contained in this Safety Data Sheet.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**8.1. Control parameters****Exposure limits**

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority

Component	European Union	The United Kingdom	France	Belgium	Spain
Nitric Acid	STEL: 1 ppm (15min) STEL: 2.6 mg/m ³ (15min)	STEL: 1 ppm 15 min STEL: 2.6 mg/m ³ 15 min	STEL / VLCT: 1 ppm. indicative limit STEL / VLCT: 2.6 mg/m ³ . indicative limit	STEL: 1 ppm 15 minuten STEL: 2.6 mg/m ³ 15 minuten	STEL / VLA-EC: 1 ppm (15 minutos). STEL / VLA-EC: 2.6 mg/m ³ (15 minutos).

Component	Italy	Germany	Portugal	The Netherlands	Finland
Nitric Acid	STEL: 1 ppm 15 minuti. Short-term	TWA: 1 ppm (8 Stunden). AGW -	STEL: 1 ppm 15 minutos	STEL: 1.3 mg/m ³ 15 minuten	TWA: 0.5 ppm 8 tunteina

	STEL: 2.6 mg/m ³ 15 minuti. Short-term	TWA: 2.6 mg/m ³ (8 Stunden). AGW -	STEL: 2.6 mg/m ³ 15 minutos TWA: 2 ppm 8 horas		TWA: 1.3 mg/m ³ 8 tunteina STEL: 1 ppm 15 minuutteina STEL: 2.6 mg/m ³ 15 minuutteina
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Component	Austria	Denmark	Switzerland	Poland	Norway
Nitric Acid	MAK-KZGW: 1 ppm 15 Minuten MAK-KZGW: 2.6 mg/m ³ 15 Minuten	STEL: 1 ppm 15 minutter STEL: 2.6 mg/m ³ 15 minutter	STEL: 2 ppm 15 Minuten STEL: 5 mg/m ³ 15 Minuten TWA: 2 ppm 8 Stunden TWA: 5 mg/m ³ 8 Stunden	STEL: 2.6 mg/m ³ 15 minutach TWA: 1.4 mg/m ³ 8 godzinach	TWA: 2 ppm 8 timer TWA: 5 mg/m ³ 8 timer STEL: 4 ppm 15 minutter. value calculated STEL: 10 mg/m ³ 15 minutter. value calculated

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Nitric Acid	STEL : 1 ppm STEL : 2.6 mg/m ³	STEL-KGVI: 1 ppm 15 minutama. STEL-KGVI: 2.6 mg/m ³ 15 minutama.	STEL: 1 ppm 15 min STEL: 2.6 mg/m ³ 15 min	STEL: 1 ppm STEL: 2.6 mg/m ³	TWA: 1 mg/m ³ 8 hodinách. Ceiling: 2.5 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Nitric Acid	STEL: 1 ppm 15 minutites. STEL: 2.6 mg/m ³ 15 minutites.	STEL: 1 ppm 15 min STEL: 2.6 mg/m ³ 15 min	STEL: 1 ppm STEL: 2.6 mg/m ³	STEL: 2.6 mg/m ³ 15 percekben. CK	STEL: 1 ppm STEL: 2.6 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Nitric Acid	STEL: 1 ppm STEL: 2.6 mg/m ³ TWA: 0.78 ppm TWA: 2 mg/m ³	STEL: 1 ppm STEL: 2.6 mg/m ³	STEL: 1 ppm 15 Minuten STEL: 2.6 mg/m ³ 15 Minuten	STEL: 1 ppm 15 minuti STEL: 2.6 mg/m ³ 15 minuti	STEL: 1 ppm 15 minute STEL: 2.6 mg/m ³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Nitric Acid	Skin notation MAC: 2 mg/m ³	Ceiling: 2.6 mg/m ³	TWA: 1 ppm 8 urah TWA: 2.6 mg/m ³ 8 urah STEL: 1 ppm 15 minutah STEL: 2.6 mg/m ³ 15 minutah	Binding STEL: 1 ppm 15 minuter Binding STEL: 2.6 mg/m ³ 15 minuter TLV: 0.5 ppm 8 timmar. NGV TLV: 1.3 mg/m ³ 8 timmar. NGV	STEL: 1 ppm 15 dakika STEL: 2.6 mg/m ³ 15 dakika

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

Derived No Effect Level (DNEL)

No information available

Predicted No Effect Concentration (PNEC)

No information available.

8.2. Exposure controls

Engineering Measures Ensure that eyewash stations and safety showers are close to the workstation location

Personal protective equipment

Eye/face Protection Wear chemical splash goggles and face shield. If splashes are likely to occur:. Goggles.

Skin and body protection Wear protective gloves/protective clothing.

Respiratory Protection No protective equipment is needed under normal use conditions. In case of inadequate ventilation wear respiratory protection.

Environmental exposure controls No information available

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES**9.1. Information on basic physical and chemical properties**

Physical State	Liquid
Appearance	Clear
Odor	None
Odor Threshold	No information available
pH	1
PH Range	0 - 2.5

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
Melting point/freezing point	No information available	
Boiling Point/Range	~ 100 °C / 212 °F	
Flash Point (High in °C)	No information available	
Evaporation Rate	No information available	
Flammability (solid, gas)	No information available	
Flammability Limit in Air		
Upper flammability limit:	No information available	
Lower flammability limit:	No information available	
Vapor pressure	No information available	
Vapor Density	No information available	
Specific Gravity	No information available	
Water Solubility	Soluble	
Solubility in other solvents	No information available	
Partition coefficient	No information available	
Autoignition Temperature	-	
Decomposition Temperature	No information available	
Kinematic viscosity	No information available	
Dynamic viscosity	No information available	
Explosive Properties	No information available	
Oxidizing Properties	No information available	

9.2. Other information

Softening Point	No information available
Molecular Weight	No information available
VOC Content(%)	No information available
Density	No Information available
Bulk Density	No information available

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

No information available

10.2. Chemical stability

Stable under normal conditions

Explosion Data

Sensitivity to Mechanical Impact None
Sensitivity to Static Discharge None

10.3. Possibility of hazardous reactions

None under normal processing

10.4. Conditions to avoid

Extremes of temperature and direct sunlight

10.5. Incompatible materials

No information available

10.6. Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapors

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

Acute Toxicity

Unknown Acute Toxicity 0 % of the mixture consists of ingredient(s) of unknown toxicity.

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (inhalation-gas) 20,833.00 ppm
ATEmix (inhalation-dust/mist) 2.17 mg/L
ATEmix (inhalation-vapor) 53.69 mg/L

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Water	LD50 > 90 mL/kg (Rat)		
Nitric Acid			LC50 = 2500 ppm (Rat) 1 h

Skin Corrosion/Irritation Causes severe burns

(b) skin corrosion/irritation; Category 1. B.

Serious eye damage/eye irritation Risk of serious damage to eyes

Sensitization No information available

Mutagenic Effects No information available

Carcinogenic effects No information available

Reproductive Effects No information available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Symptoms Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

Aspiration hazard No information available

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12. ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects

0% of the mixture consists of component(s) of unknown hazards to the aquatic environment

12.2. Persistence and degradability

Persistence

Soluble in water, Persistence is unlikely, based on information available.

12.3. Bioaccumulative potential

Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Nitric Acid	-2.3	No data available

12.4. Mobility in soil

Highly mobile in soils

Mobility

Component	log Pow
Nitric Acid 7697-37-2 (0 - 10%)	-2.3

12.5. Results of PBT and vPvB assessment

No information available

12.6. Endocrine disrupting properties

This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant

This product does not contain any known or suspected substance

Ozone Depletion Potential

This product does not contain any known or suspected substance

SECTION 13. DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

Other Information Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1 UN-No UN2031
 14.2 Proper Shipping Name NITRIC ACID
 14.3 Hazard Class 8
 14.4 Packing Group II
 Description UN2031, NITRIC ACID, 8, II
 14.5 Marine Pollutant Not Applicable
 14.6 Special Provisions None
 EmS No. F-A, S-B
 14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code No information available

ADR

14.1. UN number UN2031
 14.2. UN proper shipping name NITRIC ACID
 14.3. Transport hazard class(es) 8
 14.4. Packing group II

ICAO

14.1 UN-No UN2031
 14.2 Proper Shipping Name NITRIC ACID
 14.3 Hazard Class 8
 14.4 Packing Group II
 Description UN2031, NITRIC ACID, 8, II
 14.5 Environmental hazard Not Applicable
 14.6 Special Provisions None

IATA

14.1 UN-No UN2031
 14.2 Proper Shipping Name NITRIC ACID
 14.3 Hazard Class 8
 14.4 Packing Group II
 Description UN2031, NITRIC ACID, 8, II
 14.5 Environmental hazard Not Applicable
 14.6 Special Provisions None
 ERG Code 8L

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS), U.S.A. (TSCA).

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
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Water	7732-18-5	231-791-2	-	-	X	X	KE-35400	X	-
Nitric Acid	7697-37-2	231-714-2	-	-	X	X	KE-25911	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Water	7732-18-5	X	ACTIVE	X	-	X	X	X
Nitric Acid	7697-37-2	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

European Union

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Water	7732-18-5	-	-	-
Nitric Acid	7697-37-2	-	Use restricted. See item 75. (see link for restriction details)	-

<https://echa.europa.eu/substances-restricted-under-reach>

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

Component	Germany - Water Classification (AwSV)
Nitric Acid 7697-37-2 (0 - 10%)	WGK1

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Nitric Acid 7697-37-2 (0 - 10%)	Prohibited and Restricted Substances		

15.2. Chemical safety assessment

A Chemical safety assessment according to regulation (EC) No. 1907/2006 is not required

SECTION 16: OTHER INFORMATION**Full text of H-Statements referred to under sections 2 and 3**

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

H272 - May intensify fire; oxidizer

EUH071 - Corrosive to the respiratory tract

Key or legend to abbreviations and acronyms used in the safety data sheet

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

TWA TWA (time-weighted average)

Ceiling Maximum limit value

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

STEL STEL (Short Term Exposure Limit)

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Full text of H-Statements referred to under section 3

H314 - Causes severe skin burns and eye damage

H272 - May intensify fire; oxidizer

EUH071 - Corrosive to the respiratory tract

Prepared By	Regulatory Affairs
Prepared For	Thermo Fisher Scientific Inc.
Issue Date	No information available
Revision Date	18-Apr-2024
Reason for revision	SDS sections updated.

This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.

Disclaimer

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End of Safety Data Sheet